



Testing Population Proportions Transcript

Hello everyone, welcome to lesson 2 video 1. In this session, we will study tests for population proportions. The test helps us to analyse categorical data such as yes or no, true or false and so on. The main topics in this video include one sample and two sample proportion tests.

In our discussion so far, we have discussed or focused our studies mainly on the measurement data like height, weight, time, temperature, where we test the mean, right? But many of the most important questions we face aren't about averages. They are about percentages, proportions and so on. So now we are moving beyond averages in this entire lecture on testing population proportions or percentage of people or percentage of observations, okay? So what exactly is a proportion? While a mean summarises measurement data, a proportion summarises categorical or count data.

It is a fraction of percentage of a population that possesses a certain characteristic like so let's say for example, there is an election in India. So how many percentage or what percentage of the people in India favours a particular party? Let's say it could be Congress or BJP or XYZ, right? These are the examples. Let's say the percentage of voters in Rajasthan who support a particular candidate or a party, the proportion of manufactured parts that are defective.

Let's say you are working in a company and that company is producing hundreds of bottled water and there are 1% defective out of them. So which is 1 bottle defective out of 100 productions that you are making, right? Then the fraction of students who pass a certification exam. So these are all about proportions, percentages.

It is not about the means, right? Now similarly, let's start with the simplest case, testing a claim about a single population proportion. This is called one sample Z-test of proportion, okay? Now in one sample Z-test, we use this sort of formula. This is very similar to what we have for means, it's just that we are taking proportions into account.

So instead of having $\bar{X}$, we have P and we have the population statistic or what we state in the null hypothesis, okay? So the setup is very familiar to the framework that you already know. The null hypothesis is the true population proportion or P_0 here, right? And then we use test Z-statistics as the difference between what we have got from the sample, what we know from the population, and then the standard error at the bottom, okay? The important thing to notice here is we calculate the standard error using P_0 , the value from null hypothesis. This is because we always begin our test by assuming that the null hypothesis is true.

And then when we get enough support in favour of alternate hypothesis, then we say that the alternate hypothesis is true and null is what we fail to accept, right? So with



this consideration, let's move into looking how do we actually do the testing. For doing the test, there has to be certain criterias which meet. First of them is the sample has to be random.

This is required because we want the sample to be good representative of the entire population. We do not want the sample to not represent the population well, okay? So what we do is we randomly take samples and we take sufficiently large samples such that we accommodate for everything, okay? Then the second condition is large counts or large sample size. Now if we take a very small sample size, the estimate that we are looking for or the statistic that we are looking for might not be good enough, okay? So the sample should be large enough to expect at least some amount of successes or failures, okay? That's what it means by the product of n into p_0 is greater than 10 and the product of n into q or $1 - p_0$ is greater than or equals to 10, right? Why it is important? Because if let's say if you are testing a claim about 1% defect rate in a manufacturing setup and you have a sample of 50, 1% of that would be 0.5 defects, which is so small, right? It will be unreliable.

So the Z-test would be invalid here, okay? Now there are times when we want to test between two groups. The statistic is same. I mean we are measuring a categorical data, yes, no, true, false, high, medium, low, whatever it is, but we are measuring it against two groups.

What if I say, if I have to find out the proportion of students who pass an exam at BITS PLANY is higher or different than the proportion of students who pass at Hyderabad campus, right? So we might want to hypothesise, let's say they are exactly same, which is our null hypothesis and we'll say that they are different in our alternate hypothesis. So one way of writing null hypothesis is also, right, they are equal or the difference between them is zero, right? Similarly, the second example, which I have quoted here, does a new marketing campaign result in higher proportion of clicks than the older one. So you are comparing two different proportions.

Now the entire formulation, if you look into, it looks very similar to what we have in the cases of averages for the groups. It's just that the standard error formulation might look slightly different, but the denominator, which is the standard error, it is coming as a pooled. The reason for having pooled is we always are looking for a statistic which is common, common to both the proportions.

And our best estimate for that is to combine and pool the data. I'll give you more examples or we'll solve more examples when we'll be going through the live classes. But for the time being, consider this as a formulation, wherein the top is the difference that you have got from the two proportions or the two populations, the estimates of them and then minus zero, which is our alternate, sorry, null hypothesis, right? And at the bottom we have a standard error, okay? Now, let's move on to this interactive chart wherein we have one sample test.



We want to look into one sample test and let's say that we have got some sample size of 50. Sample proportion is what we are looking for is 0.60. Now for that, the Z statistics comes out to be 1.41 using the formula which I showed in the slide before right here. Now, similarly, the P value comes out to be 0.157. Now if you see on the bottom, I am calculating whether the sample size is good enough to represent the statistics that I have got.

Now if I change the sample size to a very low value, you see that the formulation does not give you enough large sample to test the statistics. Hence whatever you will get is always wrong. So, the first thing is to get to a level wherein the number of samples are much more than what you require to develop or do the statistics or to do the test, right? If you see when I change these values, the bottom becomes green which means that we have got sufficiently large amount of sample to test the criteria.

When we have number of samples which is lesser than 10, then it turns red, right? Similarly, this is sample proportion and based on that you see how Z statistics and the P value changes. So this is all I had for this video wherein we were discussing about there are situations when we do not want to study about averages or compare averages. We might want to compare proportion and that proportion might be coming from one population, it might be coming from two different populations.

When it is coming from one population, let's say for example someone comes up with a claim that the average mileage of cars being produced at Pune is, let's say it's high or low. So to test that statistics because the variable in study, it's a categorical variable, high or low, right? In that case, we will be using the Z or T test for proportion. We will not be using the test for averages because herein the variable which we are studying is categorical in nature.

It is not continuous, okay? And then the second case is when you have two groups and you would want to understand whether the number of females or the proportion of females, not number, proportion of females in Mumbai is greater than or lesser than the proportion of females in Delhi, right? That's also proportion or percentage. So these are the situations wherein you will be using Z test, either one sample test when the population, when the sample is coming from one population, when you are trying to make all of these a study or you are trying to understand whether two proportions are similar or different, then you will be taking two sample test. It is very similar to what we are doing with respect to one sample T test and one sample, sorry, two sample T test and one sample T test.

It's just that we are measuring now proportions and not averages, okay? In summary, test for proportions allows us to analyse categorical data and compare group differences. This makes them specifically useful in surveys, marketing experiments,



product testing, and so on. In the next video, we will study the chi-square test for independence of categorical data.

-----End-----